uTile Nano Manual

Revision 0

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1 Introduction

This Manual is for the uTile PLC system ported to run on the Arduino Nano controller board. The manual mainly describes how to get uTile up and running quickly and it should be read in conjunction with the comprehensive m328-uTile-Manual which covers the generic operation of uTile in detail.

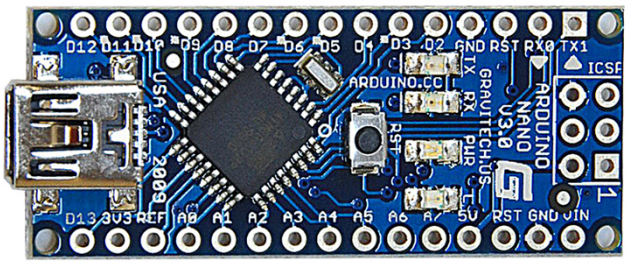
The uTile for the Nano controller board is a complete, fully functional and ready-to-run PLC that is supplied pre-programmed on an Arduino Nano controller board. You will need to provide a human user interface (HUI) to communicate with uTile's the built-in editor. The HUI for uTile can be any VT100 type terminal emulator runing on a PC or a tablet computer.

The manual shows you how to connect the Nano controller pre-programmed with uTile to the terminal emulator and to get to the uTile sign-on screen.

1.1 Hardware Parts list

You need the following parts to get started:

- Arduino Nano controller board pre-programmed with the uTile PLC in Flash program memory.
- Mini-USB cable to connect the UNO board to the PC USB port.



1.2 Software Terminal Emulator Programs

Several Open-source terminal emulator programs are available for free that can be used for uTile's HUI to provide the screen display and keyboard input for uTile's program editor. uTile communicates with a DEC VT100 type terminal emulator via the USB serial interface.

The Tera Term communications program runs under several versions of Windows OS and performs well as a HUI for uTile.

If you are running a *nix (Unix-like) OS like Linux, FreeBSD etc, the Open-source Minicom communications program works well with uTile and is usually available from the distribution's software repository.

The C-Kermit 9.0 communications program works very well as a HUI for uTile and the powerful command and script language offers the added advantage of automating the file transfer procedure that allows uTile to store and retrieve User File application programs to and from the host PC. For this reason, C-Kermit is the preferred communications program for uTile's HUI.

The Kermit 95 Windows OS equivalent version of C-Kermit is unfortunately not Open-source and therefore it is not available for free.

If you aren't able to use C-Kermit for whatever reason you can use any of the other recommended terminal emulator programs and get a fully functioning uTile HUI except for the file transfer functions. The file transfer functions are nice-to-have but are not essential for using uTile.

2 Installing Tera Term

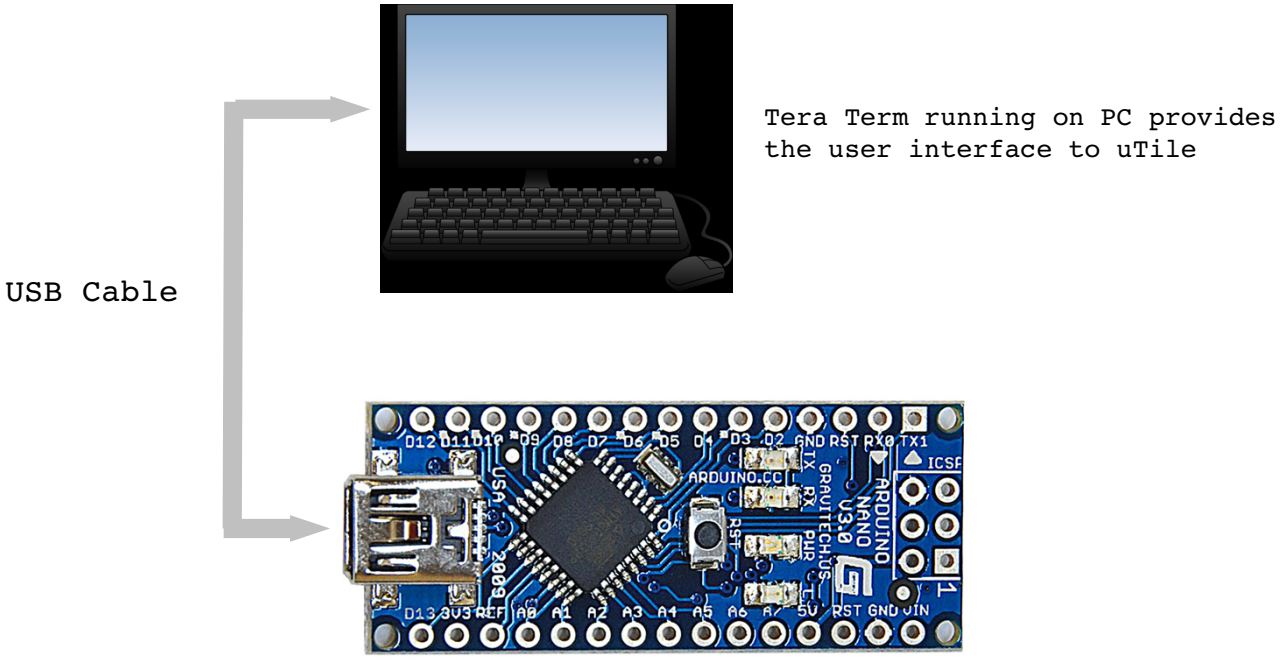
Tera Term comes under a BSD style license and runs under Windows 7 operating system.

You can download a copy of Tera Term from the following web site:

#https://ttssh2.osdn.jp/index.html.en

Install the program in a suitable directory on you PC's hard drive.

Run Tera Term to see that it is installed and operating properly. The website has manuals available for reference. This section covers the installation of Tera Term and describes how to set the communications parameters to match the uTile system.



Start Tera Term and configure it to the right settings by following the steps below.

1) Plug the USB cable from the PC back into the Nano's USB socket

2) Start Tera Term on the PC and set up the serial port connection by selecting the Setup menu tab and clicking on Serial port. Enter the serial port parameters into the boxes.

- Serial port setup parameters: -

Port:	COMx (COM port assigned to Nano).
Baud rate:	19200
Data:	8 bit
Parity:	none
Stop:	1 bit
Flow control:	none

Exit the menu then save the setup configuration from Setup menu, Save setup.

The uTile terminal screen will appear and from here you have complete access to the uTile system. The uTile system is now ready to use.

```
                  <<<<  Nano - u T I L E  -  ATmega328  >>>>
-------------------------------------------------------------------------------
          Bit Stack                                        Port Byte
       7 6 5 4 3 2 1 0              Data               7 6 5 4 3 2 1 0
       ===============              =====              ===============
       0 0 0 0 0 0 0 0              00000
-------------------------------------------------------------------------------

      000  .....         008  .....         016  .....         024  .....
      001  .....         009  .....         017  .....         025  .....
      002  .....         010  .....         018  .....         026  .....
      003  .....         011  .....         019  .....         027  .....
      004  .....         012  .....         020  .....         028  .....
      005  .....         013  .....         021  .....         029  .....
      006  .....         014  .....         022  .....         030  .....
      007  .....         015  .....         023  .....         031  .....



    Enter:
-------------------------------------------------------------------------------
 *** User File 0 ***
```

Please refer to the m328-uTile-Manual for a full technical description of uTile.

2.1 Tera Term Questions and Answers

Q1. Tera Term does not connect to the Nano, why?

A1. Make sure you have selected in Tera Term the correct serial port settings and COM port number that is mapped to the USB serial connection used by the UNO. You may need to remove and re-apply power to the UNO board, or to close and re-start Tera Term to have it rescan the serial USB port. After you establish a successful connection, you should save the setup configuration so Tera Term will continue to use the working configuration when you start it up the next time.

A2. Make sure you do not have more than one instance of Tera Term running at the same time. If the first instance has opened up the serial port and is running in the background, the second instance will not be able to connect. Close down the second instance of Tera Term.

Q2. Can I use another terminal emulator if I set the communications parameters to 19,200 baud, 8N1, same as Tera Term, say for example the terminal emulator provided with the Arduino IDE?

A1. You can use any terminal emulator provided it conforms to the VT100 or VT102 specifications. uTile uses escape control sequences for cursor positioning control and for managing character display attributes. Since Arduino's terminal emulator is not VT100 compatible, it won't work properly with uTile.

3 Installing C-Kermit

C-Kermit is available under an Open-source license and runs under *nix operating systems. You can usually get the application from your distribution's package repository.

The following description is for a Debian 8 Linux distribution. The instructions presented here assume you are familiar with the procedure for installing software packages on your system. Install the C-Kermit program package. Download the three files .mykermrc, download.cmd, upload.cmd from the web site http://www.plcchips.com/home.html. and copy them to your /home/(your-user-name)/ directory.

The file .mykermrc is an initialization file to configure kermit for use with uTile. This file sets up the USB device port to /dev/ttyUSB0, baud rate to 19200, 8N1 and other housekeeping chores, then opens the connection to the terminal interface to uTile.

The download.cmd file is a kermit script file that automates the procedure for getting a User's application program from uTile and storing an image of the UF0 contents on the host PC.

The upload.cmd file is a kermit script file that automates the procedure for sending a User's application program image stored on the host PC as a disc file back to uTile's UF0 program area.

Plug a USB cable into the PC and Nano board USB socket and start kermit by opening a terminal screen on the PC and invoking the command 'kermit' followed by a <CR> on the PC. If kermit is installed and operating properly you should see the uTile sign-on screen appear as described in Section 2 above.

Please visit the Kermit Project web site http://www.columbia.edu/kermit/index.html for further information on C-Kermit.

3.1 Basic commands for running Kermit

Kermit is a comprehensive communications package with a lot of commands but as we are only using a very small sub-set of the features we only need to know a few basic commands. When you connect the Nano controller to the PC with the USB cable and start kermit, the .mykermrc initiliazation file opens the user interface terminal to uTile and puts you at the main editor screen.

To quit the uTile editor screen and return to kermit's command line prompt type the following:

```
'^\c'
```

That is, press the 'CTL' key and '\' key at the same time, release both keys then press the 'c' key. Kermit closes the terminal interface and drops you back at the kermit command line prompt:

```
(/home/[your-user-name]) C-Kermit>
```

You can invoke script files from the command line and enter whatever other command you wish to invoke. To open the terminal interface to uTile from the kermit prompt, type the following command:

```
connect
```

uTile's main editor screen appears. You can just type 'c' instead of 'connect' for this command.

Whenever you wish to quit kermit from the kermit prompt, type 'q' followed by a <CR>. Kermit closes the USB port and exits back to the terminal screen from which it was started.

3.2 Using the DOWNLOAD.CMD script file

This script file is invoked at the kermit command line prompt to get uTile to send to the host PC an image of the contents of the user program that is currently in UF0 file space. The image is stored as a

disc file on the PC for later sending back to uTile.

uTile should be at the main screen and your application program and delay timer settings should already be entered. Escape from uTile back to the kermit command line by entering the '^\c' command. When you are at the kermit command line prompt, enter the following command:

'download myfile.hex' followed by a carriage return <CR>.

The 'myfile.hex' is any name you wish to put on the incoming file. The file is saved in Intel Hex file format. If you don't provide the script a filename, a default name of utile.hex is used. The hex file is an ascii readable file so you can view the file with a text editor, but you won't see the uTile instructions as they appear in the editor's UF0 page display. Instead what you will see is the binary data stored in the UF0 space in ascii hexadecimal format.

3.3 Using the UPLOAD.CMD script file

This script file is invoked at the kermit command line prompt to get the host PC to upload to uTile a hex file that was previously saved by the download.cmd script. The uploaded file is loaded to uTile's UF0 file space and the saved timer data is restored.

uTile should be at the main screen. Escape from uTile back to the kermit command line by entering the '^\c' command. When you are at the kermit command line prompt, enter the following command:

'upload myfile.hex' followed by a carriage return <CR>.

The 'myfile.hex' is the name of the file you wish to upload. Use the name of whatever file you wish to upload. If you don't provide the script a filename, a default name of utile.hex is used. A 'File not found!' error message is shown if the file does not exist, then the script will exit back to the kermit prompt.

If the file exists, it will be uploaded to uTile. When the file transfer is completer the message 'File read completed - '/' for menu -->' is shown. Enter a '/' and uTile's main screen will appear.

3.4 Kermit File Permissions

*nix (Unix-like) operating systems require users to belong to a file's group before access to the file is granted.

Communications programs like kermit and minicom require users to join the 'dialout' group before they are granted access to the serial device being used by the communications program, i.e. /dev/ttyUSB0, /dev/ttyACM0, /dev/ttyrfcomm0 etc.

Check if you belong to the 'dialout' group by looking at the output of the 'groups' command typed in at the command line in a terminal session. If you don't see 'dialout' in the list of groups you need to add yourself to this group by issuing the following command:

sudo usermod -a -G dialout your_user_name

Issue the 'groups' command again to confirm 'dialout' appears in the list of groups you belong to.

You may need to change the permissions for ***/dev/ttyUSB0*** with the command ***sudo chmod a+rw /dev/ttyUSB0***, and do the same for ***/dev/ttyACM0*** if you need to access this device for your selected controller board. You only need to do this once for each device.

4 Arduino Nano check-out

This section covers the check-out of uTile on the Arduino board to show that a user program can be entered and executed. Enter the following commands while in the editor mode (the bottom status line shows a message '*** User File 0 ***'

Enter the 'fill' command, by typing 'fill' followed be a <CR> (carriage return). The screen looks like this:

```
                  <<<<  Nano - u T I L E  -  ATmega328  >>>>
--------------------------------------------------------------------------------
          Bit Stack                                          Port Byte
       7 6 5 4 3 2 1 0              Data                 7 6 5 4 3 2 1 0
       ===============              =====                ===============
       0 0 0 0 0 0 0 0              00000
--------------------------------------------------------------------------------

      000  NOP           008  NOP           016  NOP           024  NOP
      001  NOP           009  NOP           017  NOP           025  NOP
      002  NOP           010  NOP           018  NOP           026  NOP
      003  NOP           011  NOP           019  NOP           027  NOP
      004  NOP           012  NOP           020  NOP           028  NOP
      005  NOP           013  NOP           021  NOP           029  NOP
      006  NOP           014  NOP           022  NOP           030  NOP
      007  NOP           015  NOP           023  NOP           031  NOP



    Enter:
--------------------------------------------------------------------------------
 *** User File 0 ***
```

Type in the following commands as shown below:

```
                  <<<<  Nano - u T I L E  -  ATmega328  >>>>
--------------------------------------------------------------------------------
          Bit Stack                                          Port Byte
       7 6 5 4 3 2 1 0              Data                 7 6 5 4 3 2 1 0
       ===============              =====                ===============
       0 0 0 0 0 0 0 0              00000
--------------------------------------------------------------------------------

      000  CLKA.         008  NOP           016  NOP           024  NOP
      001  .Y7           009  NOP           017  NOP           025  NOP
      002  KEY           010  NOP           018  NOP           026  NOP
      003  END           011  NOP           019  NOP           027  NOP
```

```
004  NOP          012  NOP          020  NOP          028  NOP
005  NOP          013  NOP          021  NOP          029  NOP
006  NOP          014  NOP          022  NOP          030  NOP
007  NOP          015  NOP          023  NOP          031  NOP



    Enter:
-------------------------------------------------------------------------------
 *** User File 0 ***
```

Enter the 'ex0' command to start executing the program and observe that the on-board LED starts flashing at a 200 ms rate. This shows uTile is functioning properly.

5 Nano pinout connections

uTile's input port ***PA*** is mapped to the Nano controller board pins as shown below:

Inputs	Port pin	Nano pin
A0	PC0	A0
A1	PC1	A1
A2	PC2	A2
A3	PC3	A3
A4	PC4	A4
A5	PC5	A5
Auto-start	PB0	Digital 8

Unconnected inputs are pulled high by internal pull-up resistors.

The output port ***PY*** is mapped to the Nano controller board pins as shown below:

Outputs	Port pin	Nano pin
Y0	PD2	Digital 2
Y1	PD3	Digital 3
Y2	PD4	Digital 4
Y3	PD5	Digital 5
Y4	PD6	Digital 6
Y5	PD7	Digital 7
Y6	PB4	Digital 12
Y7	PB5	Digital 13
Out_sense	PB1	Digital 9